Prabhat Kumar Singh . Jarvis Consulting

Data Engineer with 2 years of experience building scalable data pipelines, ETL workflows, and reporting solutions for Fortune 500 clients. Experienced in transforming large-scale datasets using Python, SQL, PySpark, Azure Databricks, Azure Data Factory, Azure Data Lake Storage Gen2, and database systems. Worked on Databricks-based analytics projects involving medallion architecture, Delta Live Tables, Delta tables, data quality checks, and automated workflow orchestration. Delivered analytics-ready datasets supporting business intelligence, enterprise reporting, financial analytics, and promotional analytics use cases. Holds a Bachelor's degree in Computer Science and a postgraduate specialization in Artificial Intelligence, with hands-on Jarvis projects in Linux monitoring, retail analytics, Spark processing, and cloud-based data engineering.

Skills

Proficient: Python, AWS, Azure, Databricks, Pyspark, Sql, Power BI

Competent: dbt, Docker, PostgreSQL, Data Modeling, ETL Pipeline Development, Machine Learning(Regression/Clustering)

Familiar: Hadoop/HDFS, GCP Dataproc, Java, Kafka, Financial Reporting/Financial Forcasting

Jarvis Projects

Project source code: https://github.com/jarviscanada/jarvis_data_eng_Singh

Linux Cluster Resource Monitoring App [GitHub]: Designed and implemented a lightweight Linux cluster monitoring system using Bash scripts and crontab to collect hardware specifications and real-time CPU and memory utilization metrics. Stored collected telemetry in a PostgreSQL database running inside Docker and developed SQL-based analytics queries to support capacity planning and performance monitoring. Automated scheduling and deployment to enable reliable, low-overhead monitoring across distributed environments.

PostgreSQL Club Booking Database System [GitHub]: Designed and implemented a normalized relational database system for a recreational club booking platform using PostgreSQL. Created schemas, constraints, and relationships using DDL statements and developed advanced SQL queries leveraging joins, aggregations, subqueries, and window functions to generate operational insights. Containerized the database environment using Docker and automated deployment with Bash scripts to improve portability and reproducibility across environments.

Retail Data Wrangling and Analytics [GitHub]: Developed a retail analytics solution to analyze customer purchasing behavior using Python, Pandas, SQLAlchemy, PostgreSQL, and Jupyter Notebook. Performed data cleaning, transformation, and exploratory analysis on transactional datasets and applied RFM segmentation and monthly sales trend analysis to generate customer-level insights. Delivered analytics-ready datasets and data-driven marketing recommendations to support customer retention strategies and campaign targeting.

Java Grep Application [GitHub]: Developed a command-line text search application in Java 8 that replicates core Linux grep functionality by recursively scanning directories for pattern-based matches using regular expressions. Implemented a functional programming approach using Lambda expressions and the Stream API to improve processing efficiency and memory usage through lazy evaluation. Integrated SLF4J for logging and Maven for dependency management, and packaged the application within a Dockerized environment to enable portable execution across systems while supporting scalable processing of large directory structures.

PySpark Data Analysis [GitHub]: Built distributed data analytics pipelines using PySpark across Databricks and Zeppelin notebook environments to analyze large-scale retail and economic datasets. In Databricks, ingested retail transaction data from PostgreSQL via JDBC and implemented end-to-end analytics with PySpark DataFrames, including monthly sales trends, placed vs. canceled orders, sales growth rates, active users, new vs. existing customer analysis, and RFM-based segmentation. In Zeppelin, analyzed World Development Indicators (WDI) data using PySpark with Hive-backed tables, applying transformations, joins, filtering, and aggregations to strengthen distributed processing workflows. Delivered scalable, business-focused insights into customer behavior and sales performance using structured Spark DataFrame APIs.

ETL pipeline for Financial Data [GitHub]: Financial Fraud Analytics ETL Pipeline [GitHub]: Designed and implemented an end-to-end fraud analytics data pipeline on Azure Databricks using Medallion Architecture (Bronze, Silver, Gold) to ingest, clean, enrich, and analyze multi-source financial transaction datasets. Integrated Azure SQL Database, Azure Data Lake Storage Gen2, PySpark, and JDBC to process transactions, card data, user profiles, merchant category codes, and fraud labels into analytics-ready Delta tables. Built curated Gold-layer datasets to support fraud trend monitoring, high-risk user identification, merchant-category fraud rate analysis, daily fraud loss tracking, and behavioral pattern changes before and after fraud events. Developed interactive Databricks dashboards and scheduled Databricks Jobs to orchestrate automated pipeline execution from ingestion to reporting refresh, enabling scalable and production-style analytics workflows.

COBOL Student Registration System [GitHub]: Built a COBOL-based Student Registration System using OpenCobolIDE and GnuCOBOL to simulate mainframe-style indexed file processing. Developed a menu-driven application to generate a student master file, insert new student records, update existing records, delete records, and query students by ID or insertion date. Implemented course-wise report generation using COBOL sorting, sequential file handling, indexed file access, record layouts, and structured procedural logic. Strengthened practical understanding of COBOL fundamentals, copybooks, file control entries, working-storage design, CRUD operations, and report generation in a mainframe-style business application.

Highlighted Projects

LangChain-FinanceGPT [GitHub]: Developed an AI-powered financial question-answering system using LangChain, OpenAI embeddings, FAISS vector search, and Flask to enable natural language querying of company annual reports. Implemented document ingestion pipelines to process large financial PDFs into vector embeddings and built a retrieval-augmented generation workflow to deliver context-aware responses for investment research and financial analysis use cases.

Kubernetes for Data Engineering [GitHub]: Designed and deployed a containerized data engineering workflow using Docker and Kubernetes to orchestrate scalable data processing services across distributed environments. Implemented infrastructure configuration for automated service deployment, container lifecycle management, and environment portability to support reliable execution of analytics workloads in cloud-native architectures.

Professional Experiences

Data Engineer, Jarvis (2026-present): Designed and delivered analytics-focused data solutions including monitoring dashboards, relational reporting systems, and Python-based ETL pipelines for enterprise reporting use cases. Developed automated data ingestion workflows and advanced SQL queries to generate performance metrics and operational insights from large-scale datasets. Collaborated within Agile teams to manage and prioritize development tickets, translating business requirements into reliable data-driven deliverables that improved reporting accuracy and visibility across stakeholder groups.

Data Engineer, Decision Point Analytics (Feb 2021-Dec 2022): Designed and delivered scalable data engineering and analytics solutions supporting Marketing Mix Modeling (MMM) and promotional analytics initiatives for Fortune 500 clients. Developed Python-based ETL pipelines and advanced SQL transformation workflows using Databricks across Azure and AWS cloud environments to process large-scale marketing and sales datasets. Automated data ingestion and validation workflows and produced analytics-ready datasets powering reporting dashboards and campaign performance measurement. Collaborated within Agile teams to translate business requirements into reliable data-driven deliverables that improved reporting accuracy, marketing attribution visibility, and decision-making across stakeholder groups.

Education

SRM University (2016-2020), Bachelor of Computer Sciences, Computer Science - First Class with Distinction

Georgian College (2023-2024), Postgraduate Certificate in Artificial Intelligence, School of Information Technology - Dean's List

Miscellaneous

- Microsoft Certified: Azure Data Fundamentals
- Databricks Certified Data Engineer Associate
- IBM Data Science Professional Certificate
- Machine Learning with Python (IBM)
- Databases and SQL for Data Science with Python (IBM)
- Introduction to the Internet of Things (IoT)- Curtin University